A-A-0051660(DM)
9 September 1986

INTERIM COMMERCIAL ITEM DESCRIPTION

MANOMETER SET, VENOUS BLOOD PRESSURE

The General Services Administration has authorized the use of this interim commercial item description which was developed by the Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA, 19101, based upon currently available technical information.

This interim commercial item description covers a sterile non-pyrogenic, disposable manometer set designed to measure central venous blood pressure either continuously or intermittently.

Salient characteristics.

General. Shall be a sterile, non-pyrogenic, disposable manometer set designed to measure central venous blood pressure continuously or intermittently by way of an intravenous injection set.

Material. The plastic tubing shall be clear, colorless, transparent, flexible polyvinyl chloride. The manometer column shall be clear, colorless, rigid, plastic. The "Y" type injection site shall be fabricated from clear or translucent rigid plastic. The diaphragm or plug at the injection site shall be fabricated from natural rubber. The plastic components coming in contact with the fluid being administered shall meet the requirements of the Pharmacopeia of the United States for Class IV plastics (USP) and for transfusion and infusion assemblies when tested in accordance with the USP. Those rubber and plastic components coming in contact with the solution being administered and end protectors shall contain not more than 5 parts per million (ppm) total lead, barium and cadmium.

AMSC N/A FSC 6515

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Characteristics. The manometer shall be calibrated to measure venous blood pressure from -2 to +38 centimeters (cm) in 1/2 cm increments. Sixty inches plus or minus 5 inches of flexible tubing shall be permanently attached in a leakproof manner to the manometer. A male Luer slip adapter shall be attached to the distal end of the tubing. A four-way stopcock shall be provided. The stopcock shall be designed to provide flow paths from a solution bottle connected to an intravenous injection set to manometer, from patient to manometer, from solution bottle to patient, and with all ports open to allow continuous measurement of blood pressure. The stopcock shall be capable of closing all ports simultaneously. A "Y" type injection site shall be provided between the stopcock and the distal end of the tubing. A female Luer lock adapter shall be provided at the stopcock inlet directly opposite the tubing outlet. An end protector shall be attached to each Luer adapter. Each end protector shall be readily removable. When placed over each Luer adapter, the end protector shall, after sterilization, maintain sterility and integrity of the entire fluid path of the device until manually removed. The top of the manometer column shall have a vented dust cover. The dust cover shall have a loop to facilitate hanging of the manometer from a hook. The manometer shall have a clamp to facilitate connection of the set to a 1-inch diameter intravenous pole. A clamp adapter shall be supplied to facilitate connection to a 3/4 inch diameter intravenous pole. The column shall have a fluid level indicator float. The float shall have a specific gravity less than that of water. The float shall be colored for good visibility.

Markings. The manometer column shall be graduated from -2 to +38 centimeters (cm) in half cm increments with each whole number division marked with the appropriate numerical value.

Instructions. An instruction sheet detailing the proper use of the manometer set shall be provided with each device.

Workmanship. The manometer set shall be free from defects which detract from its appearance or impair its serviceability.

Product Quality Level.

Visual and dimensional examination. Shall be examined to verify that design, construction and workmanship are in accordance with applicable requirements.

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Materials. The supplier shall be responsible for performing tests to assure that materials are in accordance with requirements specified.

Unit. Package (PG). One package containing ten manometer sets, each set individually packaged sterile in a suitable peel open type container capable of maintaining sterility of contents unless package is damaged or opened, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified herein. The unit of product for test purposes shall be one manometer set. AQL shall be 1.0 percent defective. Inspection level shall be S-2.

Luer fittings. The male and female Luer fittings shall meet the dimensional and performance requirements of ANSI/HIMA MD70.1.

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Freedom from pyrogens. The manometer shall be tested for freedom from pyrogens in accordance with the USP.

Sterility. The manometer shall be sterilized in accordance with the USP.

Heavy Metals extractable test. A 2-gram composite sample shall be obtained from the rubber and plastic components. The material shall be cut into small slivers, so that no thickness exceeds 3 mm, and eluted for 72 hours at 50°C, 24 hours at 70°C, 60 minutes at 121°C in one menstruum of 0.9% sodium chloride. Each gram of the material shall be covered with 5 milliliters (ml) of the menstruum. The menstrua are collected, agitated after elution and then submitted to spectographic absorption, spectrophotometeric or specific wet analysis for lead, barium, and cadmium. The metals are reported in parts per million. Thus 1.0 ppm in elute is equivalent to extracting 1.0 mg from 200 gr of test material dispersed in 1,000 ml of elution menstruum.

Alternate heavy metals extractable test. Accurately weigh a 5-gram sample for each of the metals (barium, cadmium or lead) into a 1-oz. Vycor crucible. Reduce to ash in a muffle furnace at 650°C for eight hours. Let cool to room temperature and add 15 ml of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil to 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50 ml volumeteric flask. Rinse the residue on the filter with deionised water and dilute to 50 ml. Prepare standard solutions of 1, 2 and 5 ppm for each of the metals, namely barium, cadmium, and lead. Determine the absorbances of these solutions at appropriate wave lengths using a Hollow Cathode Ba, Cd, or Pb lamp for each of the respective metals and 10X scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 303, Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standard.

Calculation:

$$\frac{A \times F \times \text{dilution}}{W} = \text{ppm barium, cadium, or lead}$$

A = Absorbance

$$F = \text{Factor} = \frac{(\text{Concentration standard in ppm Ba, Cd, or Pb})^*}{\text{absorbance}}$$

W = Sample weight in grams

* The three standards are averaged to give ppm/unit A.

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Biocompatibility. Plastic material shall be tested in accordance with the USP tests for Class IV plastics. Rubber material shall be tested in accordance with the Intramuscular and Intracutaneous sensitivity tests of the USP.

Certificates of biocompatibility. A certificate of compliance, stipulating that offered items meet the biocompatibility test, shall be acceptable, provided that the contractor has supplied acceptable items to the Government within the past three years and that there have been no changes to material formulation and method of processing since actual testing was performed. The certificate shall include the date the testing was performed, the test method and results, the name and title of the person evaluating the results, the current contract number and NSN, a statement certifying that material processing and formulation have not changed since the time that biocompatibility was performed and the signature and title of the person making the certification.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

NSN	ITEM IDENTIFICATION
6515-01-125-3251	MANOMETER SET, VENOUS BLOOD PRESSURE

Ordering data (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-3827

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